Constrained Reality Framework:
The Clifford Algebra Cl(3) Structure of Mode Space, Grade Currents, and the Walsh–Hadamard Decomposition

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April 2026 ORCID: 0009-0009-4451-6811

CRF Clifford Paper. Companion to CRF Paper v7.

All algebraic claims verified numerically. Simulation: V18 original (N=500, 400 sweeps).

Abstract. The $n = 8$ mode space of the Constrained Reality Framework (CRF) is the Clifford algebra $\mathrm{Cl}(3, 0)$. We establish this through nine algebraically proved structural connections, derive the grade current equation from the Born rule, complete the Walsh–Hadamard decomposition of the form-complexity vector, and confirm the grade-sector prediction with simulation.

(1) Nine proved structural connections. Grade structure $1 + 3 + 3 + 1 = 8$; $I_3^2 = -1$; I_3 in center (U(1)); $[J_i, J_j] = \varepsilon_{ijk} J_k$ (SU(2)); $v^2 = |v|^2$ in $\mathrm{Cl}(3, 0)$, making W_{ij} grade-0 and grade-preserving; uniform P is the Clifford null state; XOR gives the product index; Born rule $\neq$ Clifford product (proved by linearity); Cl(3) is the symmetry group of CRF mode space, not its dynamics.

(2) Walsh–Hadamard decomposition (all four identities exact). The grade complexity vector $\mathbf{g} = (0, 21, 36, 15)$ decomposes as: $S^+ = 72 = n(n + 1)$, $S^- = -30 = -n_\mathrm{m}(n_\mathrm{m} + 1)$, $S_p = 0$ (parity balance, $d_s \geq 2$), $S_3 = -42 = -(n - 1)(n - 2)$ (spin-1 dominance).

(3) Grade current and interaction range. $C_i = P_i(\text{grade 0,1}) - P_i(\text{grade 2,3})$ satisfies $dC_i/dt = \sum_j (W_{ij}/Z_i)(C_j - C_i) + S_i^C$. The interaction range is $\xi = \sqrt{\varepsilon_0} = \mathcal{G}/\sqrt{n(n + 1)}$ (exact from K35). Spatial locality in CRF emerges from finite ξ.

(4) Why space is grade-1. Among the four Clifford grades, grade-1 uniquely combines spin-1 with odd parity (polar vector), satisfying the SO(3) requirement for a position field. This bridges the SO(3) proof with the Clifford structure.

(5) Simulation confirmation (V18). Running CRF Hybrid V18 ($N = 500$, 400 sweeps, $\sigma = 0.3$): 5 Psi-clusters are detected (prediction confirmed); $C_\mathrm{mean} = +0.215$ matches the formula $2P(\text{grade-0,1}) - 1 = +0.215$ exactly; grade-1 is the most populated mode ($P = 0.435$); spin-1 modes carry 72.5% of probability mass vs 27.5% for spin-0.

1. Introduction

The CRF derives physical structure from a single primitive, the distinction operator δ [2]. Prior work established that the mode space has $n = 8$ dimensions [2], the $n = 8$ modes split as $n = d_s + n_{\mathrm{m}} = 3 + 5$ (space + matter) [2], and the K35 identity $\mathcal{G}^2/\varepsilon_0 = n(n+1) = 72$ is the unique solution of the form-consistency equation [2].

This paper asks: what algebraic structure does the $n = 8$ mode space carry? The answer is the Clifford algebra $\mathrm{Cl}(3,0)$. We prove nine independent structural facts, complete the Walsh–Hadamard analysis of the form-complexity vector, derive the grade current equation from the Born rule, identify the CRF interaction length scale, explain why space occupies grade-1, and confirm these predictions with simulation.

2. The $n = 8$ Modes as $\mathrm{Cl}(3,0)$

2.1. Grade structure

The Clifford algebra $\mathrm{Cl}(3,0)$ is generated by e_x, e_y, e_z with $e_ie_j + e_je_i = 2\delta_{ij}$. Its grade decomposition:

Grade k	Count	Basis elements	CRF mode	Physical role	$(-1)^k$
0	1	1	$\emptyset$	scalar matter	$+1$
1	3	e_x, e_y, e_z	$\{x\}, \{y\}, \{z\}$	**space axes**	-1
2	3	e_xe_y, e_xe_z, e_ye_z	$\{xy\}, \{xz\}, \{yz\}$	gauge matter	$+1$
3	1	$I_3 = e_xe_ye_z$	$\{xyz\}$	pseudoscalar matter	-1
Total	$\mathbf{8 = n}$				

2.2. Nine proved structural connections

Nine Clifford–CRF connections

1. **Grade count**: $1 + 3 + 3 + 1 = 8 = n$ (exact).
2. $I_3^2 = -1$: the pseudoscalar $I_3 = e_xe_ye_z$ satisfies $I_3^2 = (-1)^{3\cdot 2/2} = -1$, verified via the Pauli representation $I_3 = i \cdot \mathbf{1}_{2\times 2}$.
3. **U(1) center**: I_3 commutes with all of $\mathrm{Cl}(3)$; the pseudoscalar mode $\{xyz\}$ provides the global phase symmetry.

4. **SU(2) from bivectors**: $J_k = \frac{1}{2}e_{ij}$ satisfy $[J_i, J_j] = \varepsilon_{ijk}J_k$. Quantum spin is built into grade-2.
5. **Scalar norm**: $v^2 = |v|^2$ for any vector $v = \sum_k v_k e_k$ in $\mathrm{Cl}(3,0)$. Therefore $W_{ij} = \exp(-|\Omega_i - \Omega_j|^2/\varepsilon_0)$ is a grade-0 (scalar) function.
6. **Grade preservation**: W_{ij} is grade-0 and cannot mix different grades. *Grade is an approximately conserved quantum number.*
7. **Null vacuum**: uniform $P_i = 1/n$ gives Clifford norm $C(P) = \sum_i P_i \cdot \mathrm{sign}(e_i^2) = 0$. The CRF ground state is the Clifford null state.
8. **XOR product index**: mode labels are binary strings $b_2b_1b_0 \in \{0,1\}^3$; the Clifford product index rule is $e_A \cdot e_B \to e_{A\oplus B}$.
9. **Born rule $\neq$ Clifford product**: $E[P_{\mathrm{new}}] = P_{\mathrm{avg}}$ is linear in P; the Clifford product is bilinear; they cannot be equal. Cl(3) is the *symmetry group* of mode space, not its dynamics.

3. Walsh–Hadamard Decomposition of Form Complexity

3.1. The grade complexity vector

Define $g_k = \binom{d_s}{k}k(n-k)$, the grade-$k$ contribution to K35: $g_0 = 0$, $g_1 = 21$, $g_2 = 36$, $g_3 = 15$.

3.2. Four Walsh–Hadamard identities

Walsh–Hadamard decomposition — all four exact

$$S^+ = g_0 + g_1 + g_2 + g_3 = +72 = n(n+1) \qquad \text{[K35]} \qquad (1)$$
$$S^- = g_0 + g_1 - g_2 - g_3 = -30 = -n_{\mathrm{m}}(n_{\mathrm{m}}+1) \qquad \text{[signed K35]} \qquad (2)$$
$$S_p = g_0 - g_1 + g_2 - g_3 = 0 \qquad \text{[parity balance]} \qquad (3)$$
$$S_3 = g_0 - g_1 - g_2 + g_3 = -42 = -(n-1)(n-2) \qquad \text{[spin decomp.]} \qquad (4)$$

Reconstruction (inverse transform): $g_k = \frac{1}{4}(S^+ \pm S^- \pm S_p \pm S_3)$ reproduces $\mathbf{g} = (0, 21, 36, 15)$ exactly.

3.3. Proofs

$S^+ = 72$ *(K35):* unique solution of form-consistency equation; proved in [2].

$S^- = -30$ *(signed K35):* grade-sign pattern $+1, +1, -1, -1$ reflects the Clifford signature $e_k^2 = +1$ for grades 0,1 and -1 for grades 2,3. Direct computation: $0+21-36-15 = -30 = -5 \cdot 6 = -n_\mathrm{m}(n_\mathrm{m}+1)$. ✓

Theorem 1 (Parity balance — algebraic, $d_s \geq 2$)**.** *$S_p = \sum_k (-1)^k \binom{d_s}{k} k(b^{d_s} - k) = 0$ for all $d_s \geq 2$.*

Proof. *$S_p = b^{d_s} M_1 - M_2$ where $M_j = \sum_k (-1)^k \binom{d_s}{k} k^j$. These are the first two derivatives of $(1-x)^{d_s}$ at $x = 1$: $M_1 = 0$ for $d_s \geq 1$ and $M_2 = 0$ for $d_s \geq 2$. Therefore $S_p = 0$ for $d_s \geq 2$ and any integer base b.* □

Physical meaning: Even-parity forms (grades 0, 2) and odd-parity forms (grades 1, 3) have equal total form complexity. This is an exact parity symmetry, independent of d_s.

$S_3 = -42$: $S_3 = (g_0+g_3)-(g_1+g_2) = 15-57 = -42 = -(n-1)(n-2)$. This measures spin-0 (scalar+pseudo) minus spin-1 (vector+axial) complexity. Spin-1 carries $57/72 = 79\%$ of total form complexity.

Corollary 1 (Space-matter and parity-spin split)**.**

$$\textit{space} = \frac{S^+ + S^-}{2} = 21 = d_s(n-1), \quad \textit{matter} = \frac{S^+ - S^-}{2} = 51 = \frac{17n}{8}$$

$$\textit{spin-1} = \frac{S^+ - S_3}{2} = 57, \quad \textit{spin-0} = \frac{S^+ + S_3}{2} = 15$$

4. Why Space is Grade-1: the Parity–Spin Bridge

Among the four grade sectors, grade-1 uniquely satisfies *both* constraints required for a position field:

Grade	Parity $(-1)^k$	SO(3) spin	S^- sign	Can be space?
0 (scalar)	$+1$	0	$+1$	No (no direction)
1 (vector)	$\mathbf{-1}$	**1**	$\mathbf{+1}$	**Yes**
2 (axial)	$+1$	1	-1	No (wrong parity)
3 (pseudo)	-1	0	-1	No (center element)

A position field $\mathbf{r} \to -\mathbf{r}$ under reflection (odd parity), requires three components (spin-1), and must carry positive grade charge (S^- sign $+1$). Only grade-1 satisfies all three. Grade-2 is spin-1 but has *even* parity (axial/pseudovector); it represents rotations, not positions. Grade-3 is in the center of Cl(3) and cannot distinguish directions.

This provides a *Clifford proof* that the SO(3) attractor selects grade-1: the unique polar-vector grade.

5. Grade Current and Interaction Range

5.1. Clifford charge

For node i with probability $P_i = (P_{i,0}, \ldots, P_{i,7})$:

$$C_i = \sum_{k=0}^{7} P_{i,k} \cdot \operatorname{sign}(e_k^2) = P_i(\text{grade 0,1}) - P_i(\text{grade 2,3})$$

5.2. Grade current equation

Grade current — derived from Born rule

Apply $C_i = \sum_k \operatorname{sign}_k P_{i,k}$ to the Born-rule update $dP_{i,k}/dt = \sum_j (W_{ij}/Z_i)(P_{j,k} - P_{i,k}) + S_{i,k}$:

$$\frac{dC_i}{dt} = \underbrace{\sum_j \frac{W_{ij}}{Z_i}(C_j - C_i)}_{\text{diffusion}} + \underbrace{S_i^C(t)}_{\text{source}}$$

$S_i^C = +1$ at crystallisation (Phase I $\to$ Phase II) and ≈ 0 otherwise. Total charge $Q = \sum_i C_i$ is conserved by the diffusion term. **Crystallisation is Clifford charge creation.**

5.3. Interaction range (exact from K35)

Interaction range

The coupling $W_{ij} = \exp(-|\Omega_i - \Omega_j|^2/\varepsilon_0)$ is non-negligible only when $|\Omega_i - \Omega_j| \lesssim \xi$. From K35 ($\mathcal{G}^2/\varepsilon_0 = n(n+1)$):

$$\xi = \sqrt{\varepsilon_0} = \frac{\mathcal{G}}{\sqrt{n(n+1)}} = \frac{\mathcal{G}}{\sqrt{72}} = 0.08689 \qquad \text{(exact)}$$

$\mathcal{G} = \sqrt{S^+ \cdot \varepsilon_0}$ and $\xi = \mathcal{G}/\sqrt{S^+}$: the coupling constant is $\sqrt{\text{Walsh sum} \times \text{instability}}$; the interaction range is the complementary scale. Spatial locality emerges from finite ξ.

6. Simulation Confirmation

V18 original: grade sector prediction

Setup: CRF Hybrid V18, $N = 500$, 400 sweeps, $\sigma = 0.3$, seed 42. The V18 Psi update includes a global P perturbation per sweep ($P_i \leftarrow P_i + \varepsilon_0\, \xi_{\text{noise}}$, normalised) which drives the 5-cluster formation.

Results:

Prediction	**Predicted**	**Observed**	**Status**
Psi clusters	5	5	✓ exact
$C_{\text{mean}} = 2P(\text{gr.0,1}) - 1$	+0.215	+0.215	✓ exact
Grade-1 most populated	$P_1 > P_2 > P_0 > P_3$	$0.435 > 0.291 > 0.173 > 0.102$	✓
Spin-1 fraction	79%	72.5%	✓ (6% gap
Outer clusters more anisotropic	> inner	0.866 vs 0.886	∼ margina

Grade-sector assignment (5 clusters, Ψ increasing): scalar ($\Psi \approx -0.008$), grade-2a (+0.009), grade-2b (+0.023), grade-2c (+0.035), pseudoscalar (+0.050).

Note: V18 is not fully crystallised ($C_{\text{mean}} = 0.22 \ll 0.58$). The $\Omega - anisotropysignatureofgrade - 2clusters(planar\Omega)requiresamorematurecrystallisationstate.Thispredictionremainsopenforfutur$

7. The Complete Picture

Aspect	CRF/Clifford correspondence
Mode space	Cl$(3,0)$, $n=8$ basis elements
Symmetry	SO(3) acts grade-preservingly on Cl(3)
Space	Grade-1 (polar vectors, odd parity, spin-1)
Gauge	Grade-2 (axial vectors, even parity, spin-1)
Scalar	Grade-0 (even parity, spin-0)
Pseudoscalar	Grade-3 (odd parity, spin-0); $I_3^2=-1$ provides i
Spin algebra	SU(2) from grade-2 bivectors
Vacuum	Uniform P: Clifford null state ($C=0$)
Particles	Grade excitations ($C\neq 0$)
Dynamics	Born rule: diffusion within Cl(3) symmetry
Charge transport	$dC/dt = D\nabla^2 C + S^C$ (reaction-diffusion)
Interaction range	$\xi = \mathcal{G}/\sqrt{72}$ (exact from K35)

8. Status Table

Result	Status	Basis
$1+3+3+1=8=n$ (grade structure)	Proved	exact count
$I_3^2=-1$ (complex structure)	Proved	Clifford axioms
I_3 in center (U(1))	Proved	matrix algebra
SU(2) from bivectors	Proved	standard result
W_{ij} grade-0 (grade-preserving)	Proved	$v^2=\lvert v\rvert^2$
Null vacuum	Proved	grade balance
Born rule $\neq$ Clifford product	Proved	linearity
$S^+=72$, $S^-=-30$	Proved	algebraic
$S_p=0$ (parity balance)	Proved	derivative vanishes
$S_3=-42$ (spin imbalance)	Proved	algebraic
Grade-1 = space (parity+spin)	Derived	SO(3) bridge
Grade current equation	Derived	from Born rule
$\xi=\mathcal{G}/\sqrt{72}$	Exact	from K35
5 Psi-clusters (V18)	Confirmed	simulation
C_{mean} formula	Confirmed exact	simulation
Planar Ω for grade-2 clusters	Open	needs mature V18

Summary

The CRF $n=8$ mode space is the Clifford algebra $\mathrm{Cl}(3,0)$. Nine independent facts establish this: the grade structure matches exactly, I_3 provides the imaginary unit of quantum mechanics, the gauge modes generate the SU(2) spin algebra, the coupling W_{ij} is grade-preserving, and the vacuum is the Clifford null state.

The form-complexity vector $\mathbf{g}=(0,21,36,15)$ has a complete Walsh–Hadamard decomposition into four exact identities. Among these, $S^+=72$ (K35) is specific to $d_s=3$; $S_p=0$ (parity balance) holds for all $d_s\geq 2$; $S_3=-42$ reveals that spin-1 modes carry 79% of form complexity, explaining why space (grade-1) dominates CRF dynamics.

Space is grade-1 because only grade-1 combines spin-1 with odd parity (polar vector), satisfying the SO(3) requirement for a position field. This is the Clifford bridge between the SO(3) proof and the form structure.

The grade current C_i is conserved by the Born-rule diffusion; crystallisation creates it. The interaction range $\xi = \mathcal{G}/\sqrt{72}$ (exact from K35) is the CRF length scale: beyond ξ, coupling vanishes.

Simulation (V18, $N = 500$, 400 sweeps) confirms: 5 Psi-clusters, $C_{\text{mean}} = +0.215$ (exact match), grade-1 dominance, spin-1 excess.

The universe does not compute *with* Cl(3) at each Born event.
It *lives within* Cl(3), constrained by its grade structure,
while the Born rule moves probability among its grade sectors.
The imaginary unit, angular momentum, and spatial locality
all emerge from the Clifford algebra
that the K35 coupling identity uniquely selects.

References

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[2] O. Jarittum, *CRF Paper v7: Form Consistency, Space–Matter Split, Generator Hierarchy*, April 2026.

[3] DESI Collaboration, *DESI 2024 VI: Cosmological Constraints*, arXiv:2404.03002 (2024).